

Fruit Battery

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Familiar objects make electricity

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 1 — Build a Fruit Battery

Hook: Familiar objects make electricity.

Learning outcomes

By the end of this session, students should be able to:

- Build a simple electrochemical cell from fruit and metal electrodes
- Explain oxidation and reduction in a galvanic cell
- Identify anode and cathode and describe electron vs. ion flow
- Measure voltage with a multimeter
- Explain why cells in series add voltage
- Describe why batteries eventually die (internal resistance, reactant depletion)

Session plan (90 min)

Time	Activity	Notes
0–10 min	Hook: measure one lemon cell; ask whether a lemon can make electricity	See lecture.md — Opening hook
10–25 min	Concept: zinc oxidation, copper cathode, acid electrolyte, electron flow	See lecture.md
25–45 min	Build one cell; measure voltage	See experiment.md — Part 1
45–60 min	Series battery challenge	See experiment.md — Part 2
60–75 min	Power LED, buzzer, or clock; discuss V, I, and internal resistance	See experiment.md — Part 3
75–85 min	Side experiments: fruits, metals, pH, salt, parallel	See experiment.md — Extensions
85–90 min	Start copper solution prep for Session 2	See preparation.md

Key reactions

Anode: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^{-}$

Cathode: $2\text{H}^{+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{H}_2(\text{g})$

Overall: $\text{Zn(s)} + 2\text{H}^{+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$

Files in this folder

File	Purpose
lecture.md	~20 min concept block
experiment.md	Hands-on procedures and data
materials.md	Session checklist
preparation.md	Pre-class and end-of-session prep

Instructor reminders

- Roll lemons/limes first to break internal membranes
- Cells in series add voltage ~linearly; current stays limited
- Galvanized nail = steel + zinc coating (zinc is the active metal)
- pH near cathode may become less acidic (protons consumed)

Status

- ☐ Lecture outline finalized
- ☐ Experiment procedure tested
- ☐ Materials procured
- ☐ Copper solution prep started for Session 2

Session 1 — Preparation

Before class (instructor)

- [] Procure fresh fruit (firm, not dried out)
- [] Test one cell: expected ~0.8–1.0 V for lemon + Zn + Cu
- [] Verify multimeters have fresh batteries and work on DC voltage
- [] Pre-cut copper strips if students struggle with stiff wire
- [] Prepare waste collection for fruit and electrodes

Day-of setup (15 min before)

- [] One station per group: fruit, electrodes, wires, multimeter, load device
- [] Optional extension materials at a side table
- [] Vinegar/salt/copper jars ready for end-of-session prep

During class — end of session (85–90 min)

Copper solution for Session 2

Recipe (per jar):

1. ~100 mL white vinegar
2. 1–2 teaspoons table salt; stir to dissolve
3. Add clean copper (gaskets, coiled wire) — maximize surface area
4. Cover loosely (not airtight) — air helps oxidize copper
5. Label with date and group name

Expected result: Pale blue, green, or blue-green solution after overnight standing. Color intensity varies.

Backup for Session 2: Prepare **0.5 M CuSO₄** (e.g. 125 g CuSO₄·5H₂O per liter, or 25 g per 200 mL). See Session 2 experiment for the full recipe table.

After class

- [] Store copper prep jars in ventilated area
- [] Note which jars look most promising for Session 2
- [] Dispose of fruit waste appropriately

Open questions / notes

- Typical voltage achieved with our materials:
- Best fruit type in pilot:
- LED lighting requires ___ cells in series:

Session 1 — Materials

Per group (3–4 students)

- ☐ 3–6 lemons, limes, potatoes, apples, or oranges
- ☐ Galvanized nails or zinc strips (2+)
- ☐ Copper strips, copper wire, or copper coins/pennies (2+)
- ☐ Alligator clip wires (4–6)
- ☐ Multimeter
- ☐ LED, small buzzer, or low-power digital clock
- ☐ pH paper or pH indicator strips
- ☐ Paper towels

Optional (extensions)

- ☐ Graphite pencil leads
- ☐ Aluminum foil
- ☐ Steel nails
- ☐ Magnesium ribbon
- ☐ Table salt (small amount for salt-addition test)

End-of-session prep (Session 2 setup)

- ☐ White vinegar (~100 mL per jar)
- ☐ Table salt (1–2 tsp per jar)
- ☐ Copper gaskets, strips, or wire (large surface area)
- ☐ Jam jars or beakers with loose lids
- ☐ Labels and marker

Instructor / room

- ☐ Waste bin for used fruit (not food waste)
- ☐ Hand-washing access
- ☐ Demo multimeter on projector or large display (optional)

Quantities to estimate

Item	Groups	Total
Fruit (minimum)	× 4 each	
Multimeters	1 per group	
9 V batteries	—	Not needed this session

Session 1 — Experiment: Fruit Battery

Main experiment: Lemon/lime/potato battery — generating electricity from chemistry.

Instructors: expected voltages, LED failure modes, and extension interpretation — instructor.md.

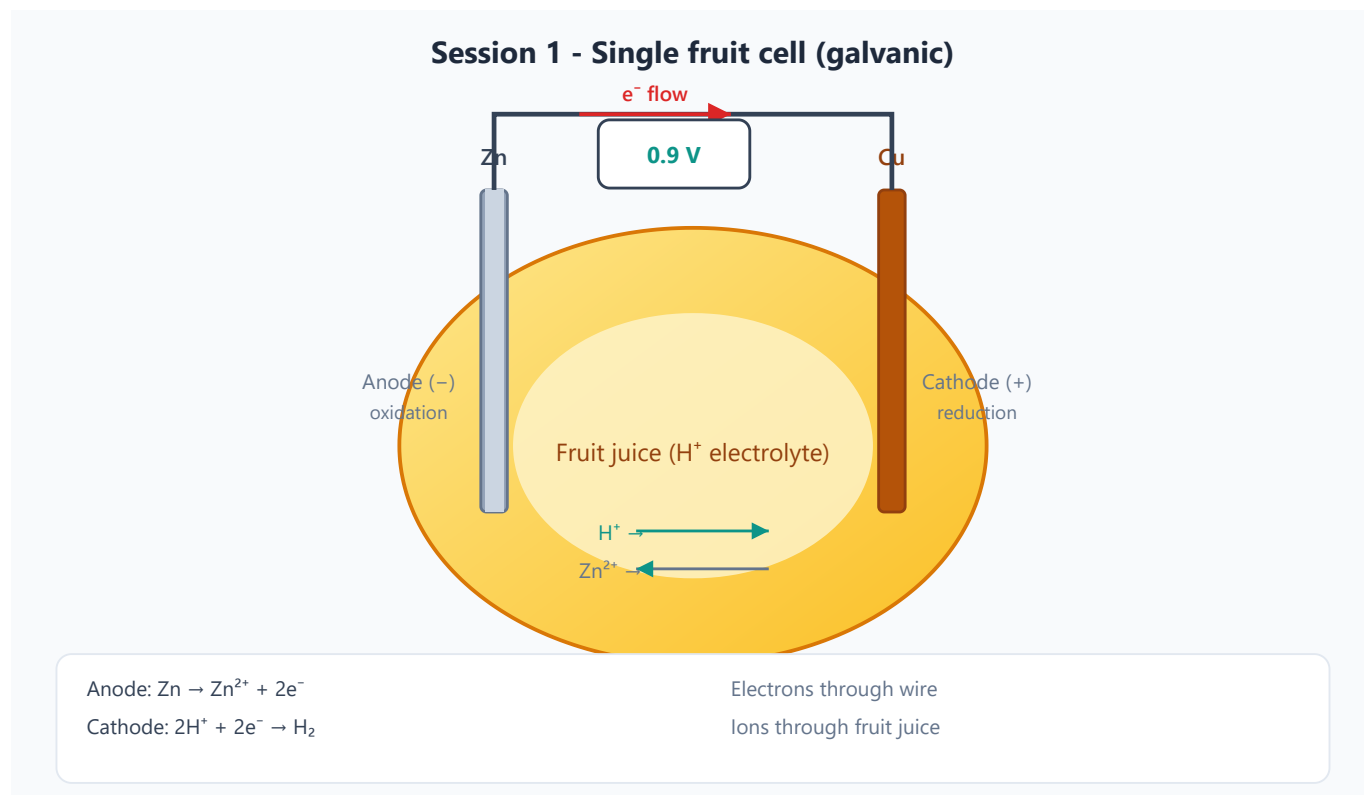


Figure 1 — Single galvanic cell: Zn anode (-), Cu cathode (+), electrons through the wire, ions through the fruit juice.

Materials checklist (per group)

Item	Spec / notes
Fruit	3–6 firm lemons, limes, potatoes, or apples
Zinc electrode	Galvanized nail or zinc strip (2+)
Copper electrode	Copper strip, stiff wire, or clean penny (2+)
Multimeter	DC voltage (and current if available)
Alligator clips	4–6 leads
Load	Red LED (1.8–2.2 V forward) and/or 1.5–3 V buzzer
Optional	pH paper, table salt, aluminum foil

Part 1 — Build one cell (25–45 min)

Procedure

1. Choose one fruit. **Roll** a citrus fruit firmly on the bench for 10–15 s (breaks juice sacs). For potato/apple, squeeze gently or make a shallow slit for better contact.
2. Insert the **zinc** electrode (galvanized nail or zinc strip) about halfway in — do not push all the way through.
3. Insert the **copper** electrode **2–3 cm away** from the zinc. Electrodes must **not touch** inside the fruit (touching shorts the cell → ~0 V).
4. Set the multimeter to **DC voltage** (V $\overline{=}$), typically the 2 V or 20 V range.
5. Clip **red (V Ω mA)** lead to **copper**; clip **black (COM)** to **zinc**.
6. Expected: copper reads **positive** relative to zinc (~0.8–1.0 V for lemon).
7. Record the open-circuit voltage. Wait 2 minutes; record again (note drift).

Expected results

Fruit (typical)	Open-circuit voltage
Lemon / lime	~0.8–1.0 V
Potato / apple	~0.5–0.9 V (varies)

Observations to record

Quantity	Value	Notes
Fruit type		
Electrode materials	Zn: ___ / Cu: ___	
Electrode separation (cm)		
Open-circuit voltage (V)		
Voltage after 2 min (V)		
Meter polarity (which metal +?)		

Troubleshooting

Problem	Likely cause	Fix
~0 V	Electrodes touching	Separate inside the fruit
Very low V	Dry fruit, poor contact	Roll fruit; re-insert; try fresher fruit
Unstable reading	Loose clips	Secure alligator clips on bare metal
Negative voltage	Leads swapped	Swap clips or note polarity and continue

Part 2 — Series battery challenge (45–60 min)

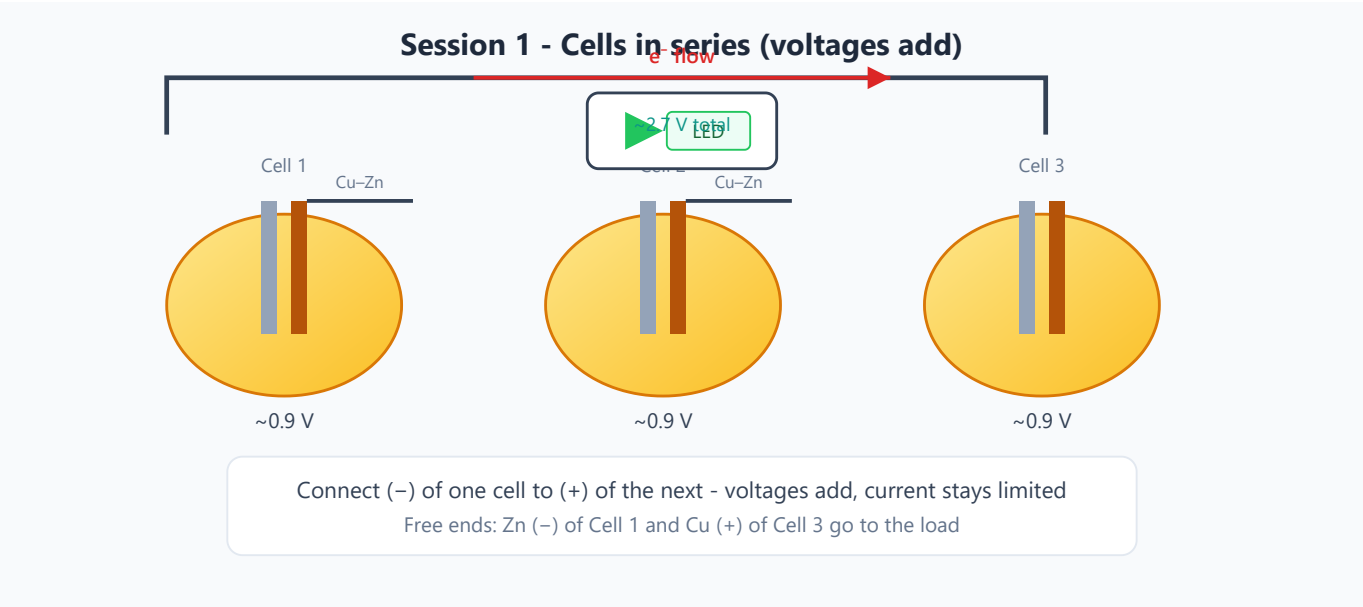


Figure 2 — Cells in series: connect Cu of one cell to Zn of the next; voltages add (~0.9 V each).

Procedure

- 1. Build **2–3** identical cells (one fruit each) the same way as Part 1.
- 2. Measure and record each cell’s voltage alone first.
- 3. Wire **in series**:
- 4. Copper of cell 1 → zinc of cell 2 (alligator lead)
- 5. Copper of cell 2 → zinc of cell 3
- 6. Measure total voltage between the **free copper** (positive end) and **free zinc** (negative end).
- 7. Compare measured total to the sum of the individual voltages.

Challenge question

"If each cell is 0.9 V, how many cells do you need for a 3 V buzzer? For a 1.5 V clock?"

Data table

# cells in series	V ₁	V ₂	V ₃	V _{total} measured	V _{sum} individual	Match?
1		—	—			
2			—			
3						

Expected

3 lemon cells ≈ **2.4–3.0 V** open circuit (enough in voltage for many red LEDs; current may still be limiting).

Part 3 — Power something (60–75 min)

Load options (try in order)

1. Multimeter on **DC mA** current range (small load — should work)
2. **Red LED** — longer leg / flat side conventions: longer leg usually **+** (to copper end of series string). Needs ~2 V+ and a few mA. Use **3 cells** first.
3. Active buzzer rated **1.5–3 V DC** (with leads)
4. Low-voltage potato/lemon clock kit (if available)

LED wiring reminder

- Series string: free **copper** → LED **anode (+)**; LED **cathode (–)** → free **zinc**
- If it does not light, reverse the LED once before concluding failure

Measurements

Load	# cells	Voltage under load (V)	Current (mA)	Works?
Open circuit			—	
LED				
Buzzer/clock				

Discussion after attempt

- Did voltage **drop** when connected to a load? (Internal resistance)
 - Why might the LED need more **current** than the fruit cell can supply?
-

Extensions — Side experiments (75–85 min)

Students choose one or more:

A. Compare fruits

Same Zn/Cu electrodes, different fruits → compare open-circuit voltage.

B. Compare metals

Same fruit; replace Zn with aluminum foil, steel nail, or Mg (if available) vs. copper. Record V and note any oxide-film issues (especially Al).

C. pH changes

After 10+ min under load, touch pH paper near each electrode. Hypothesis: cathode region may become less acidic.

D. Salt addition

Add a **pinch** (~0.25 tsp) of table salt into a slit in the fruit; re-measure V and current under the same load.

E. Parallel vs. series

Two cells in parallel (Cu–Cu and Zn–Zn tied): voltage $\approx$ one cell; compare current capability to series if meters allow.

Extension report template

Variable changed	Control	Result	Explanation (hypothesis)

End-of-session task — Copper solution prep for Session 2 (85–90 min)

Start overnight prep so Session 2 has Cu^{2+} solution ready.

Recipe (per jar)

Ingredient	Amount
White vinegar	~100 mL
Table salt	1–2 tsp (~5–10 g); stir to dissolve
Clean copper	Gasket, coiled wire, or strips — maximize surface area

Tap water is fine if you dilute vinegar slightly; vinegar itself is the main solvent. No need for distilled water here.

Steps

1. Mix vinegar + salt in a jam jar.
2. Add copper metal so much of it is submerged.
3. Cover **loosely** (not airtight — air helps dissolve copper).
4. Label with group name and date.
5. Leave overnight in a ventilated area.

Expected next day: pale blue, green, or blue-green liquid. Instructor also prepares **0.5 M CuSO_4** backup (see Session 2 experiment).

Student handout — Quick reference

Build: Roll fruit $\rightarrow$ Zn + Cu, not touching $\rightarrow$ measure V

Series: Cu of one cell to Zn of the next $\rightarrow$ voltages add

LED: Needs ~3 cells + correct polarity; voltage may sag under load

Safety: Do not eat experimental fruit; wash hands after metals

Experiment status

- [] Procedure pilot-tested with actual fruit/electrodes
- [] Typical voltage range documented (record: ___ V)

- ☐ LED/buzzer compatibility verified (# cells needed: ____)
- ☐ Extension options ranked by time available
- ☐ Copper prep jars staged for end of class

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution	No saltwater; no flames near setup; instructor-only pop test
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H₂/O₂ container